Adrien Hélias, PhD (he/him)

DATA SCIENTIST | QUANTITATIVE RESEARCHER

Quantitative Analysis - Statistical Modelling - Python - SQL

Email: adrien.helias44@gmail.com
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Website: ahelias-astro.github.io
Publications: 3 lead, 2 co-author articles

TECHNICAL EXPERTISE

Programming: Python - SQL - LaTeX - Matlab - Git - Linux - Bash - HTML (basic) - C++ (basic)
Data Analysis: Quantitative analysis - Databases - Statistical modelling - Data visualization - Machine Learning
Tools: Microsoft Office Suite - LLMs - Adobe Illustrator & Photoshop - Da Vinci Resolve - Overleaf - TOPCAT
Communication Skills: Public speaking - Scientific writing - Scientific Outreach
Languages: English - French (native)

PROFESSIONAL EXPERIENCE

DOCTORAL RESEARCHER - *WESTERN UNIVERSITY / LONDON, ON, CANADA / **2022-2026***

- **Developed and released a Python package (*ocean*)** implementing a new wavelet analysis technique for quantifying variability in irregularly sampled astronomical time series across multiple timescales
- **Designed and executed quantitative analyses** of complex astronomical datasets, applying advanced statistical methods on time series (CARMA model fitting, Slepian Wavelet Variance), and machine learning techniques (MCMC) to extract meaningful information and quantify uncertainty
- **Created a research project from scratch** based on a personal idea and **mentored a MSc student** who led the project successfully to publication
- **Communicated results to specialist and non-specialist audiences** through peer-reviewed publications and presentations at national and international scientific conferences

SCIENCE COMMUNICATOR - *CRONYN OBSERVATORY / LONDON, ON, CANADA / **2023-2026***

- **Organized** weekly events where I explained astronomy to the general public and school groups of all ages
- **Led** hands-on activities, astronomy demos, and solar/nighttime stargazing with the telescopes

GRADUATE RESEARCHER - *QUEEN'S UNIVERSITY / KINGSTON, ON, CANADA / **2022***
*MCMASTER UNIVERSITY / HAMILTON, ON, CANADA / **2021***

- **Performed statistical modelling** of the gas kinematics in dwarf galaxies
- **Analyzed large datasets of galaxies** and identified connections between physical parameters
- **Collaborated with a software developer** on the creation of simulated datasets
- **Presented findings** at a national scientific conference

EDUCATION

PHD, ASTRONOMY - *WESTERN UNIVERSITY / CANADA / **2022-2026***

MSC, ASTRONOMY - *OBSERVATOIRE DE PARIS / FRANCE / **2021-2022***

MSC, FUNDAMENTAL PHYSICS - *UNIVERSITÉ PARIS-SACLAY / FRANCE / **2019-2022***

BSC, PHYSICS - *UNIVERSITÉ PARIS-SACLAY / FRANCE / **2017-2020***

AWARDS

Teaching Assistant Award of Excellence - *Department of Physics & Astronomy, Western University /* **2026**
Paul & Kathryn Lycett Graduate Scholarship - *Department of Physics & Astronomy, Western University /* **2024**